

Linear algebra: Exercise session 2

Morteza Rahimi

Autumn 2025

-
1. Consider following matrices:

$$A = \begin{bmatrix} 1 & -2 & 1 \\ 2 & 1 & 0 \end{bmatrix}, B = \begin{bmatrix} 3 & 1 & 1 \\ -2 & 4 & -1 \\ -3 & 1 & 3 \end{bmatrix}, C = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, D = \begin{bmatrix} -2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \end{bmatrix}, E = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix},$$
$$F = \begin{bmatrix} 3 & 3 & 0 \\ 0 & -3 & 1 \\ 0 & 0 & 3 \end{bmatrix}, G = \begin{bmatrix} 0 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 0 & 0 \end{bmatrix}, H = \begin{bmatrix} -4 & 1 & 2 \\ 1 & -3 & -1 \\ 2 & -1 & 7 \end{bmatrix}, I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, E_{11} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, v = \begin{bmatrix} 1 \\ -2 \end{bmatrix}.$$

- (a) Determine the type of the following matrices (square, rectangular, diagonal, upper triangular, lower triangular, zero, identity, scalar, elementary, symmetric, (strictly) diagonally dominant).
- (b) Calculate $A^T A$ and $B^T \pm B$.
- (c) Calculate AB and BA , if it is possible.
- (d) Calculate $(2H + B - G)^T$, $A^T v$, A^2 , F^2 , and D^{10} , if it is possible.
- (e) Calculate $\text{tr}((-2B^T + 4H)^T)$.

-
2. Let A, B be a $n \times n$ matrix. Compute the following expressions:

$$(I + A)^2, \quad (I + A^T)(I + A) \quad (I - A)(I + A), \quad (A + B)^2, \quad (I + A + A^2 + \dots + A^m)(I - A).$$

-
3. Write a Python program that asks the user to enter a vector representing the diagonal entries of a matrix. The program should construct and display the diagonal matrix, as well as the zero matrix and the identity matrix of the same size.
-
4. Write a Python program that asks the user to enter a matrix, then displays the matrix along with its size, transpose, and trace. The program should also check whether the matrix is symmetric and whether it is (strictly) diagonally dominant.
-
5. Write a Python program that asks the user to enter a matrix, a vector, and a scalar. The program should display the matrix, the vector, and the results of the scalar-matrix multiplication and the vector-matrix multiplication.
-
6. Write a Python program that asks the user to enter two matrices of the same size. The program should display their sum and their product (matrix-matrix multiplication).
-